

YU CAO

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Github:<https://github.com/Yucao42>, Webpage:<https://yucao42.github.io/>

EDUCATION

New York University, Courant Institute of Mathematical Sciences

Sep. 2018-Dec. 2021

Master in Computer Science

Beihang University

Aug. 2011-Jul. 2018

Master and Bachelor in Instrumentation Science and Engineering

PUBLICATION AND TECH REPORTS

- Cheng Tan, Zhichao Li, Jian Zhang, **Yu Cao**, Sikai Qi, Zherui Liu, Yibo Zhu, Chuanxiong Guo, Serving DNN Models with Multi-instance GPUs: A Case of the Reconfigurable Machine Scheduling Problem, *Arxiv 2021*
- Jianwen Jiang, **Yu Cao**, Lin Song, Shiwei Zhang, Yunkai Li, Ziyao Xu, Qian Wu, Chuang Gan, Chi Zhang, Gang Yu, Human centric spatio-temporal action localization *ActivityNet Workshop on CVPR 2018*
- **Yu Cao**, Fuqiang Zhou, Minimal Non-linear Camera Pose Estimation Method Using Lines for SLAM Applications. *IEEE Winter Conference on Applications of Computer Vision 2018*
- Fuqiang Zhou, **Yu Cao**, Xinming Wang, Fast and Resource-efficient Hardware Implementation of Modified Line Segment Detector. *IEEE Transactions on Circuits and Systems for Video Technology*, Co-First-Author

EXPERIENCE

Senior Machine Learning/Software Engineer, Bloomberg AI Finance

Feb. 2022 - present

Municipal Bond Pricing (May 2024 - present)

- Delivered thorough model reviews to senior stakeholders across quant, product, AI/application engineering, and data — covering market landscape, pricing algorithms, data, evaluations, trade-offs, and system design.
- Delivered benchmark curve model improvement end-to-end: elicited quality requirements from domain experts, derived quantitative evaluation metrics, hyperparameter-tuned the ML model, and shipped to production.
- Designed the backtesting architecture and organized cross-team execution.
- Drove cross-team delivery of a pricing-quality dashboard and reusable metrics library; artifacts directly addressed client concerns on pricing quality.
- Built an bond credit rating classification pipeline from scratch: data sourcing, EDA, splitting, modeling, evaluation, error analysis, and hyperparameter tuning, achieving production-grade accuracy.

Realtime Corporate Bond Pricing using Machine Learning (Feb. 2022 - May 2024)

- Worked across the ML lifecycle for IBVAL real-time fixed-income pricing covering $\approx 45k$ bonds every 15 seconds.
- Researched into a ML-based bond similarity model, improving cold-start pricing accuracy for illiquid bonds.
- Built the first automated Argo deployment pipeline for AI Finance client-facing services; delivered the IBVAL pricing rollout end-to-end.

Research Intern, ByteDance Applied Machine Learning Systems

May. 2021- Aug. 2021

- Researched into scheduling algorithms(greedy and Monte Carlo Tree Search) to allocate GPU resources for DNN inference jobs to save up to 20% GPUs with Multi-Instance-GPU(MIG) feature.
- Researched into scheduling problems in pipeline DNN model serving with MIG by fitting the allocation scheduling problem into continuous optimization framework using gradient descent.

Applied Scientist Intern, AWS AI Recognition, Amazon

June. 2019- Aug. 2019

- Built a multi-language scene-text localization model based on MaskRCNN detection model.
- Researched into using visual and language semantic embeddings extracted from pretrained unsupervised Auto-encoder to improve detection performance.

Algorithm Developer Intern, Face++, Megvii Inc.

Nov. 2017- Jul. 2018

- CVPR 2018 ActivityNet Challenge of Spatio-temporal Action Localization(1st place)
 - Combined I3D deep learning model with Non-local module to express Actor-Target-Relationship in videos.

AWARDS

- McCracken Fellowship by NYU Courant Institute of Mathematical Sciences

2018-2021

SKILLS

- **Programming:** Python, Pytorch, Pandas, Go, C/C++, Pytest, Linux, VHDL, Verilog
- **Distributed System:** Kafka, KServe, Kubernetes, Argo, Hera, Grafana, AWS S3
- **Machine Learning:** Deep learning, Random Forest, GBDT, XGBoost, Time Series, Attention, Transformer
- **Quantitative Finance:** Fixed Income Bond Pricing, Backtesting, Pricing Evaluation, Municipal Bonds, Illiquid Bond Pricing, Option Pricing, Bond Similarity Modeling, Pricing Curves, Machine Learning for Credit Rating